Kurtleigh Martin

BSc Information Technology Student | Aspiring Software Developer / Systems Analyst

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PROFESSIONAL PROFILE

BSc Information Technology student with practical academic experience across desktop, mobile and web development. Recent team-project work on the BeefTech mobile data platform provided hands-on exposure to Kotlin Android development, Room/SQLCipher persistence, offline synchronisation, WorkManager, REST API integration, database migrations, Git collaboration and debugging. Comfortable learning new tools, researching solutions and contributing within a team while still developing technical depth. Previous accounting and administrative experience adds accuracy, communication, record-keeping and data-management skills. Seeking internship, graduate or junior IT opportunities where these skills can continue to develop.

TARGET ROLES

Suitable for: Software Developer Intern / Graduate Developer, Systems or Business Analyst Intern, Application Support Intern, Database Intern, IT Support Graduate and Technology Trainee roles.

TECHNICAL SKILLS

Programming Languages: C#, Kotlin, C++, Python, Java, SQL
Android & Mobile Exposure: Android Studio, Kotlin, Room, WorkManager, SQLCipher, GPS/location permissions, .NET MAUI
Backend & Integration Exposure: REST APIs, Ktor/OkHttp, basic JWT-based request flows, Gradle, JSON, environment-based configuration
Databases: Oracle, SQL Server / Azure SQL, Room / SQLite, relational database design, stored procedures, data validation
Web & Desktop: ASP.NET Web Forms, C# Windows Forms, HTML, CSS, CRUD functionality, object-oriented programming
Tools: Visual Studio, Android Studio, SQL Server Management Studio, Git/GitHub, Microsoft Office, Cisco Packet Tracer, basic Render deployment exposure

EDUCATION

North-West University | BSc in Information Technology | 2024 - 2026 (Current)
- Relevant focus areas include programming, databases, software development, systems analysis, networking and IT project work.
- Academic projects have provided practical experience in designing solutions, writing and debugging code, working with databases and collaborating in development teams.

North-West University | Bachelor of Education | 2022 - 2024
- Previous tertiary studies strengthened communication, planning, presentation and learning-support skills.

Orion Secondary School | Grade 12 | 2021
- Subjects included Mathematics, Business Studies, Computer Application Technology, Accounting and Life Orientation.

WORK EXPERIENCE

Ithemba NPO | Accounting and Administrative Assistant | 2025
- Managed professional email communication and donor correspondence, and supported organised document handling.
- Assisted with accounts payable/receivable and month-end activities while maintaining accurate financial records.
- Reconciled income and expenses and maintained donor/program databases, strengthening data accuracy and record management.
- Worked with confidentiality, reliability and attention to detail in a finance and administration environment.

ACADEMIC PROJECT EXPERIENCE

BeefTech Mobile Data Platform | Kotlin, Android, Room, SQLCipher, WorkManager, Ktor/OkHttp, Gradle, Git/GitHub | *Academic Team Project*

- Contributed to a multi-module university team project for livestock/farm data management, with most of my work focused on Android database integration, offline-to-online synchronisation, farmer registration and project integration.
- Worked with the Room + SQLCipher local database foundation and the PendingSync queue used to hold operations when the device could not immediately reach the backend. Practised using DAO/repository patterns, sync status updates, retry handling and encrypted local storage.
- Helped apply the offline-sync pattern to calf registration, farmer registration and animal-movement flows: records could be saved locally first, queued for synchronisation and later sent to the backend when connectivity became available.
- Worked with WorkManager-based background and scheduled synchronisation, including morning/evening sync logic. Resolved dependency and compilation issues around CoroutineWorker, WorkerParameters and retry behaviour, and verified successful module builds.
- Contributed to the farmer-registration data flow from registration screens/session data through repositories and DAOs into the local database. Also worked with automatic GPS coordinate capture, map/location permissions and saving location information.
- Practised Android-to-backend integration using Ktor/OkHttp, REST endpoints, JSON and JWT-authenticated requests. Worked with both local emulator backend configuration and a hosted backend setup, including basic health/deployment and environment-based configuration.
- Contributed to backend/integration work for farmer registration notifications, structured JSON output and configurable email delivery. Helped test changes so notification problems would not incorrectly mark an otherwise successful registration as unsynchronised.
- Helped implement and test a user-scoped synchronisation/security policy with staged warnings for unsynchronised local data and a Day-7 account-lock/wipe rule. The work included audit logging and safeguards so already-synchronised backend records were not removed.
- Worked with Room schema changes and migration tests, including changes needed for sync/security data. This gave practical exposure to preserving existing device data instead of relying only on destructive database migration.
- Used Git/GitHub during team integration: worked on the sync-status-backend branch, pulled shared changes, merged updates from main, resolved database-related conflicts and validated clean builds/checks after integration.
- Created/used a small demo application wrapper to exercise the integrated farm-traceability and farmer-registration flows without changing the main architecture unnecessarily. Also contributed an automatic sunlight-readability UI mode for outdoor use.
- The project strengthened my practical understanding of layered Android architecture (UI/session -> repository -> DAO -> Room), offline-first design, background work, API integration, testing, debugging and collaborative source control. I am continuing to build depth in these areas as a student.

OTHER SELECTED IT PROJECTS

Eco-Tourism Management Web Application | ASP.NET Web Forms, C#, SQL / Azure SQL

- Developed academic/team web-form functionality for tours, guides, destinations and reports; worked with SQL tables, stored procedures, database connectivity and navigation.

Cross-Platform Mobile Application | C# and .NET MAUI

- Built an academic mobile application using C# and .NET MAUI, practising object-oriented design, screen structure, user input and application logic.

Library Management System | C#, Windows Forms, database concepts

- Built a Windows Forms academic application and practised event handling, validation, record management, user-input handling and debugging.

Network Design Simulation | Cisco Packet Tracer

- Configured and tested a simulated network using IP addressing, subnetting, VLAN concepts and connectivity checks.

PERSONAL PROJECT

Spark - Grade 9 Mathematics Learning Application | Kotlin, Android Studio

- Developing an Android learning application to help Grade 9 learners strengthen Mathematics skills while practising Kotlin, Android UI and application logic.

TRANSFERABLE PROFESSIONAL SKILLS

Communication • Teamwork • Time management • Analytical thinking • Adaptability • Attention to detail • Research and problem solving

LANGUAGES

English; Afrikaans

REFERENCES

Available on request.